

TYPESCRIPT

Why Types Matter

Planrs Academy | Grade 6 Session

LUNCH ONLY

Imagine a Box that can hold ANYTHING!

Think about a magical box with no rules. You could put toys, books, snacks, or even cricket balls inside!

But wait... that gets confusing, right? If you're looking for your lunch and find a dirty cricket ball instead, that's a problem!

What if we put a label on our box?

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Now everyone knows exactly what belongs inside—no more mix-ups!

JavaScript is like a busy Local Market!

Imagine a crowded bazaar with no labels on anything. You ask for sugar, but get salt! In coding, this is what happens when we don't use labels.

The Problem: $10 + "10" = "1010"?$

In JavaScript, if you try to add a number and a word, it just joins them together like a train. It doesn't know you wanted to do Math!

JavaScript gets confused without labels. It's like mixing up your Math notebook with your Hindi notebook—everything gets jumbled!

How can we fix this? Meet TypeScript!

TypeScript adds **labels** to our boxes. Just like labeling your tiffin box helps you know what's inside, TypeScript adds labels to your code.

Catch mistakes before cooking!

Before you run your code, TypeScript checks if everything is correct. It's like a helpful friend who spots mistakes early!

Static Typing: Like a Reliance Fresh store where every shelf has a clear label.

Dynamic Typing: Like a market stall where everything is in a pile.

TypeScript "Types"

Types are just labels that tell the computer what kind of data to expect.

1. **string:** For text.

"Delhi", "Zara", "Route 42A"

2. **number:** For math.

23, 100, 5.5

3. **boolean:** For Yes/No.

true (Holiday!), false (School Day)

```
let city: string = "Mumbai";  
let marks: number = 95;  
let isSunny: boolean = true;
```

Meet the Transpiler

Browsers (like Chrome) are very smart, but they only speak **JavaScript**. They don't understand TypeScript directly.

Think of it like translating from **Hindi to English**. The meaning stays the same, but now the browser can understand it.

The **Transpiler** takes our labeled TypeScript code and turns it into clean JavaScript for the browser to read!

Interactive Activity!

Can you guess the type for these labels?

1. Aadhaar Number (12 digits) → _____

2. Your Name → _____

3. "Is it raining?" → _____

RECAP

- ✓ Catch mistakes early
- ✓ Keep code clean and safe
- ✓ Labels help us stay organized!